

Introduction:

Python is a widely used programming language today. Python was developed by Guido van Rossum, a Dutch programmer, and released in 1991. The name is inspired from a BBC comedy show Monty Python's Flying Circus. Python is a successor of the ABC programming language. At the time of this writing, Python 3 is the latest major release of Python (on your computers, you may notice the program python3). Van Rossum is Python's principal author, and his continuing central role in deciding the direction of Python is reflected in the title given to him by the Python community. He is the "Benevolent Dictator For Life" (BDFL), which means he continues to oversee Python development and retains the final say in disputes or arguments arising within the community.



Figure 1: Guido van Rossum—Creator and BDFL of Python Programming Language. (Image courtesy of Wikipedia.org)

The first ever version of Python (i.e., Python 1.0) was introduced in 1991. Since its inception and introduction of Version 1, the evolution of Python has reached up to Version 3.x (till 202). Python 2.0 was released on 16 October 2000 and had many major new features, including a cycle-detecting garbage collector and support for Unicode. With this release, the development process became more transparent and community-backed. Python 3.0 (initially called Python 3000 or py3k) was released on 3 December 2008 after a long testing period. It is a major revision of the language that is not completely backward-compatible with previous versions.

Since the last 20 years, Python has been in the Top 10 most popular programming languages. At the time of this writing (July 2022), Python is the most popular language, surpassing C and Java (according to the TIOBE index).

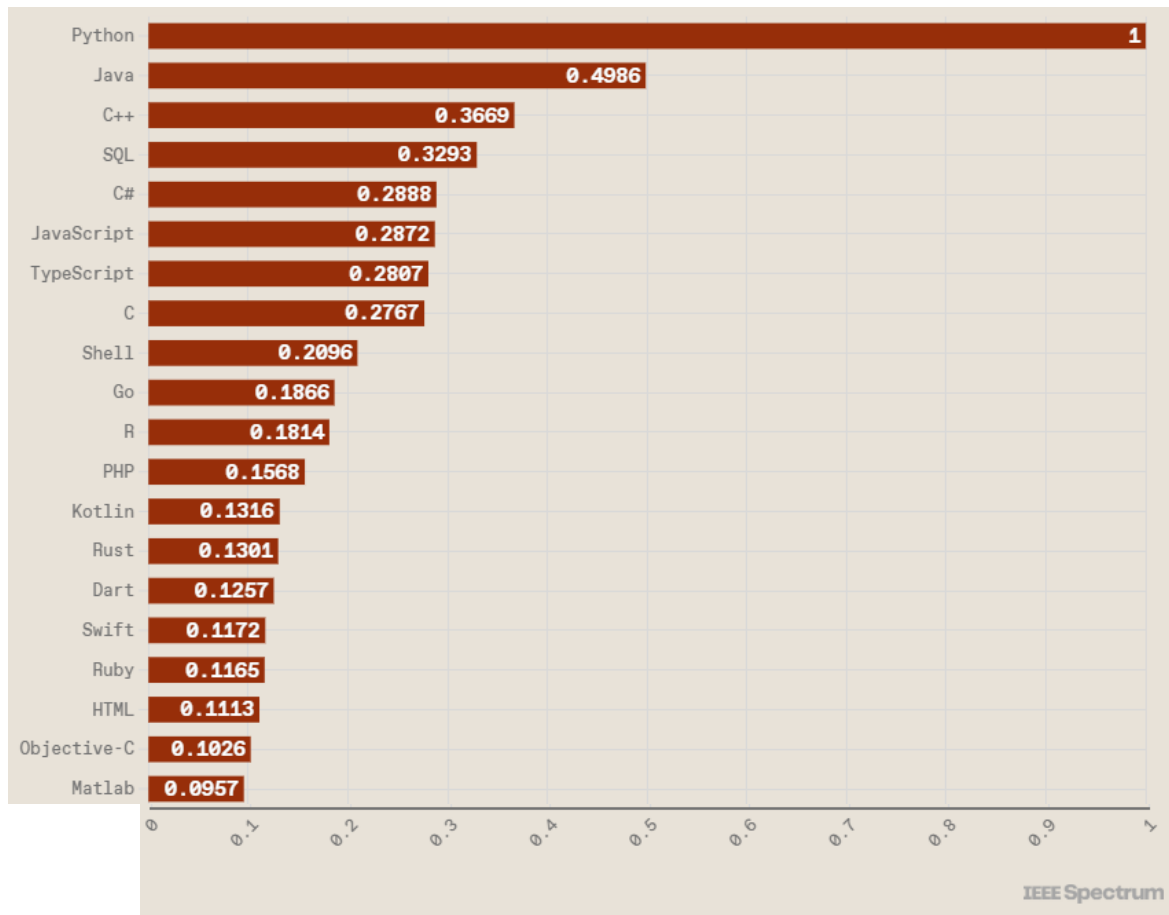


Figure 2: Ranking of programming languages by IEEE. (Image courtesy of IEEE Spectrum, New York.)

The Core Domains of Python:

1. Data Science & Advanced Analytics

Python is the industry standard for processing and visualizing data.

- Scientific Computing: Libraries like NumPy and SciPy handle complex mathematical functions and linear algebra.
- Data Manipulation: Pandas is the go-to tool for managing structured data through its powerful "DataFrame" objects.
- Statistical Modeling: Statsmodels allows for deep statistical testing and time-series analysis, often used in economics and engineering.
- Visualization: Matplotlib provides the foundation for creating professional-grade charts and graphs.

2. Web Development & Network Operations

Python simplifies the creation of both massive platforms and lightweight tools.

- Frameworks: Developers choose between Django (a robust, "all-inclusive" framework for high-security sites) and Flask (a flexible "micro-framework" for custom, smaller builds).
- API Interactions: The Requests library is the gold standard for "human-friendly" HTTP communication (handling GET/POST requests).

- Database Integration: Tools like SQLAlchemy (an ORM) allow Python to talk to databases like MySQL or PostgreSQL using Python objects instead of raw SQL code.

3. Enterprise & Cloud Infrastructure

Beyond simple scripts, Python manages the backbone of modern computing.

- Cloud Orchestration: OpenStack, written in Python, is used to build and manage private and public cloud environments.
- Platform Support: Major providers like AWS, Google App Engine, and Azure offer top-tier support for Python-based deployment.

4. Development Ecosystem

The language is supported by a mature set of tools that boost productivity.

- Integrated Environments (IDEs): Professional developers often use PyCharm for its advanced refactoring, while Spyder is preferred for scientific research.
- Pre-packaged Distributions: To avoid compatibility headaches, Anaconda and Enthought Canopy provide a single installation containing the Python interpreter and hundreds of pre-tested scientific libraries.

5. Proven Industry Success

The "**Core Domains of Python** " are best demonstrated by the global companies that rely on them:

- Social & Media: Instagram and Pinterest use Python frameworks to scale to millions of users.
- Security & Finance: PayPal and eBay utilize Python's secure libraries for high-stakes financial transactions.
- Research & Aerospace: NASA and Boeing implement Python for critical data analysis and engineering simulations.

Why to use Python:

The following are the primary factors to use python in day-to-day life:

1. Python is object-oriented

Structure supports such concepts as polymorphism, operation overloading and multiple inheritance.

2. Indentation

- Indentation is one of the greatest feature in python

3. It's free (open source)

- Downloading python and installing python is free and easy

4. It's Powerful

- Dynamic typing

- Built-in types and tools
- Library utilities
- Third party utilities (e.g. Numeric, NumPy, sciPy)
- Automatic memory management

5. It's Portable

- Python runs virtually every major platform used today
- As long as you have a compatible python interpreter installed, python programs will run in exactly the same manner, irrespective of platform.

6. It's easy to use and learn

- No intermediate compile
- Python Programs are compiled automatically to an intermediate form called byte code, which the interpreter then reads.
- This gives python the development speed of an interpreter without the performance loss inherent in purely interpreted languages.
- Structure and syntax are pretty intuitive and easy to grasp.

7. Interpreted Language

Python is processed at runtime by python Interpreter

8. Interactive Programming Language

Users can interact with the python interpreter directly for writing the programs

9. Straight forward syntax

The formation of python syntax is simple and straight forward which also makes it popular.

Python Installation:

1. There are many interpreters available freely to run Python scripts like IDLE (Integrated Development Environment) which is installed when you install the python software from <http://python.org/downloads/Steps> to be followed and remembered:

Step 1: Select Version of Python to Install.

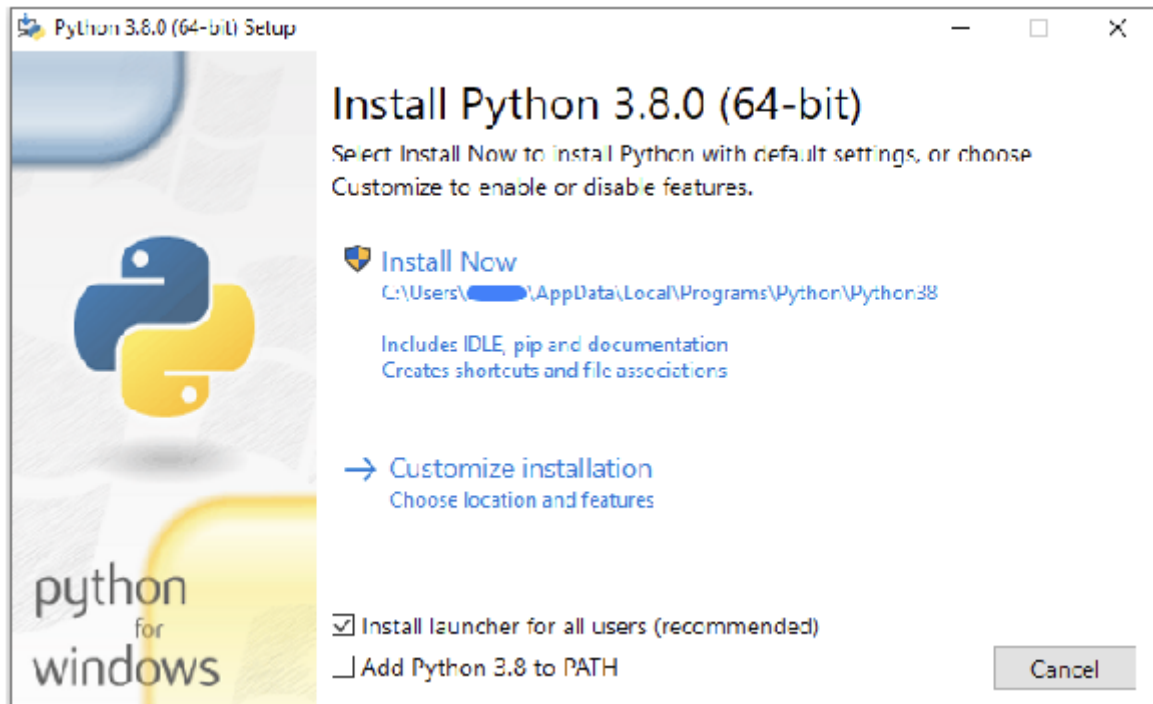
Step 2: Download Python Executable Installer.

Step 3: Run Executable Installer.

Step 4: Verify Python Was Installed On Windows.

Step 5: Verify Pip Was Installed.

Step 6: Add Python Path to Environment Variables (Optional)



2. Installing Anaconda Python Distribution:

Anaconda is a free and open source distribution of the Python programming language for data science and machine-learning related applications such as large-scale data processing, predictive analytics, and scientific computing, that aims to simplify package management and deployment. Package versions are managed by the package management system conda, which makes it quite simple to install, run, and update complex data science and machine learning software libraries like Scikit-learn, PyTorch, TensorFlow, and SciPy.

The steps described here work on the Windows 10 OS.

Step 1: Go to the link <https://www.continuum.io/downloads>. You have the option to download the 32-bit or the 64-bit version of either Python 2.7 or Python 3.6 supported Anaconda distribution. At the time of writing this book, Anaconda supported Python 3.6 version. As and when a new version of Python is released, Anaconda distribution will be updated to newer releases. In this course 64-bit Anaconda distribution supporting Python 3.6 is used to execute programs, so download the same version.

Step 2: Click on the executable file of Anaconda Python distribution which you have downloaded and the setup screen will start loading.

Step 3: You will get a welcome screen. Click on Next button.

Step 4: You will get a License Agreement Screen, read the licensing terms and click on I Agree button.

Step 5: Assuming that you are the only user on your system, select Just Me radio button. Click on Next button.

Step 6: You need to choose a location to install Anaconda Python distribution. The default installation will be under Users folder. Change the destination folder to C:\Anaconda3 to install Anaconda and click on Next button.

Step 7: In the Advanced Installation Options screen, select all the check boxes. Ignore the warnings and click on Install button.

Step 8: This starts the installation of Anaconda Python Distribution and once the Installation is complete, click on Next button.

Step 9: Finish the setup by clicking on Finish button.

Step 10: To check whether the installation is working properly or not, go to the command prompt and type python. You should see a series of lines and a prompt. This is Python Interactive mode. Here, the three greater-than signs ">>>" is the primary prompt of the interactive mode.

Other Integrated Development Environments (IDE) are PyCharm and Visual Studio.
Open source web application: Jupyter Notebook